

Particle Physics

A domain for data science — Chapter 7

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Part 1 — A domain, and the universe it studies

Why spend a week on physics?

- ▶ Data science = **statistics** + **coding** + a **domain**
- ▶ You will walk into data you don't understand, next to a **domain expert**
 - ▶ Your job: help them get the answer. Theirs: know what the answer means
- ▶ This week is practice at that: data that doesn't make sense to you yet
- ▶ **You are not tested on particle physics.** You do need to know what *“look for this signal over this background”* means

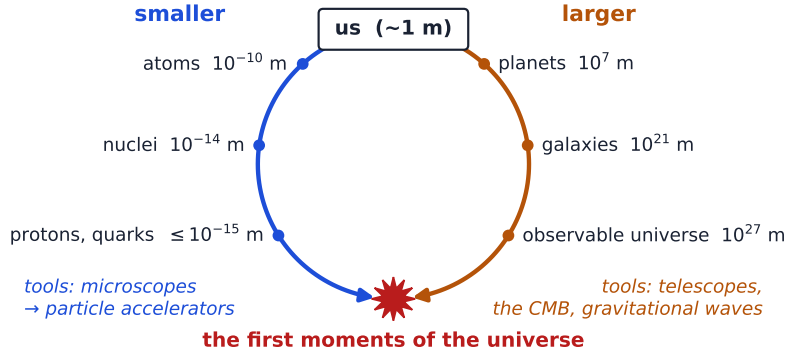
Why *this* domain

- ▶ Particle physics has lived at the **data frontier** for decades
 - ▶ Big-data problems solved with far worse hardware, software and knowledge
 - ▶ Some of it came back out: the **Web** (CERN, 1989), the computing grid, silicon pixel sensors
- ▶ The data you'll use (Lab 7, then the machine-learning labs) is the **largest dataset** in this course
- ▶ Same ideas, different words: *signal efficiency* = **true positive rate**

How to read something you only half understand

- ▶ You'll read things this week (and in any new job) that go over your head. That's normal
- ▶ Read for the **shape** first: what is measured, what is predicted, what is compared
- ▶ Write down the words you don't know; most are names for simple ideas
- ▶ Ask: *what would the data look like if the claim were true? if it were false?*
- ▶ Come back. Understanding arrives on the second or third pass

Two directions



► Small and large meet at the **beginning of the universe**

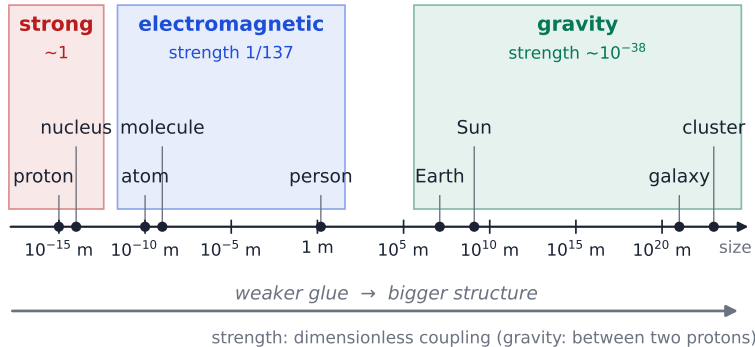
Build a universe

You're designing a universe. What do you need?

- ▶ **Space and time** to put things in
- ▶ **Stuff**: a small set of building blocks
- ▶ **Glue**: forces that hold the stuff together into structures

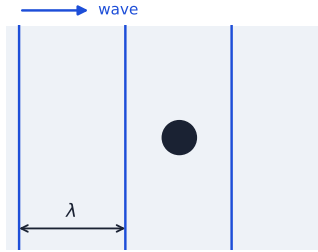
Like a house: the pieces (wood, nails) and how strongly they hold set what you can build.

Forces as glue

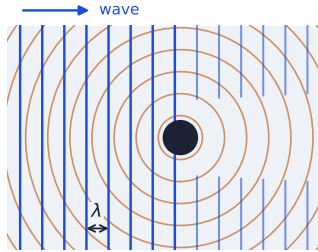


- ▶ **Strong** $\rightarrow$ nuclei · **Electromagnetic** $\rightarrow$ atoms, chemistry, us · **Gravity**: weakest, but only **adds**
- ▶ *The weaker the glue, the larger the structure*

How do you see something small?



wavelength $\gg$ object:
the wave passes by, nothing to see



wavelength $\lesssim$ object:
the wave scatters, the object shows up

to see something small you need a short wavelength, and short wavelength means high momentum, so high energy ($\lambda = h/p$)

- ▶ A wave much **longer** than the object passes it by; a **comparable or shorter** wave bounces off
- ▶ Radio passes through walls; visible light (~ 500 nm) shows a cell, not an atom

Smaller means higher energy

- ▶ Particles are waves too (de Broglie): $\lambda = h/p$, so short wavelength = high momentum
- ▶ $\lambda \approx 1.24 \mu\text{m} / (E \text{ in eV})$
 - ▶ Visible light, 2 eV $\rightarrow$ 600 nm (cells)
 - ▶ 10 keV electrons $\rightarrow$ 0.1 nm (atoms: electron microscopes)
 - ▶ 1 GeV $\rightarrow 10^{-15}$ m (a proton)
 - ▶ 1 TeV $\rightarrow 10^{-18}$ m (inside the proton)
- ▶ **Accelerators are microscopes.** Higher energy, finer detail

Fundamental, or made of something?

- ▶ Rutherford (1911): fire alpha particles at gold foil
 - ▶ Expected them to pass through a smooth “fruitcake” atom
 - ▶ Some bounced straight back: a tiny, heavy **nucleus**
- ▶ 1960s: fire electrons at protons — they scatter off hard points inside: **quarks**
- ▶ Same method every time: hit it harder, see what’s inside

$E = mc^2$: making new particles

- ▶ Energy can become **mass**: a collision with enough energy creates particles that weren't there
- ▶ Heavy particles are unstable and **decay** into lighter ones
- ▶ By the 1960s: *hundreds* of new particles — a “zoo”
- ▶ Order came from the **quark model**: the zoo is a few quarks in different combinations

The Standard Model

three generations of matter			force carriers	Higgs	
I	II	III			
quarks	+2/3 u up 2.16 MeV	+2/3 c charm 1.273 GeV	+2/3 t top 172.57 GeV	0 g gluon 0	0 H Higgs boson 125.20 GeV
	-1/3 d down 4.70 MeV	-1/3 s strange 93.5 MeV	-1/3 b bottom 4.183 GeV	0 γ photon 0	
leptons	-1 e electron 0.511 MeV	-1 μ muon 105.66 MeV	-1 τ tau 1.777 GeV	0 Z Z boson 91.19 GeV	charge symbol name mass
	0 ν_e e neutrino < 0.8 eV	0 ν_μ μ neutrino < 0.8 eV	0 ν_τ τ neutrino < 0.8 eV	±1 W W boson 80.37 GeV	

masses: PDG 2024

- ▶ 6 quarks + 6 leptons in three generations; force carriers g , γ , Z , W ; the **Higgs**
- ▶ You and everything around you: **up**, **down**, **electron**. The rest are heavier copies

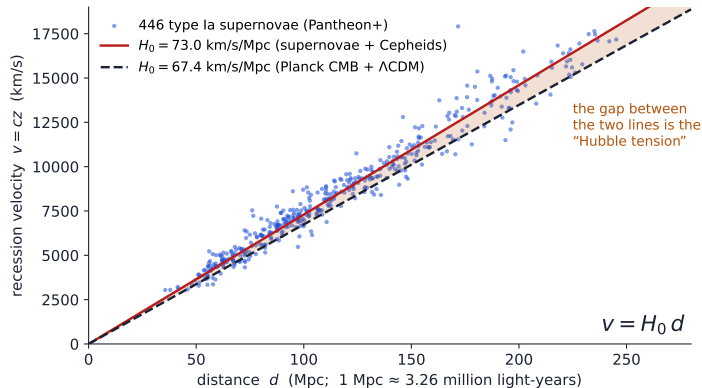
Antimatter

- ▶ Every particle has an **antiparticle**: same mass, opposite charge
- ▶ Matter + antimatter $\rightarrow$ energy. Energy $\rightarrow$ matter + antimatter, in **equal** amounts
- ▶ Antimatter is routine: made in every collider, used in hospital PET scans
- ▶ **Open question:** the early universe made equal amounts, yet today there is only matter. Why?

Looking out is looking back

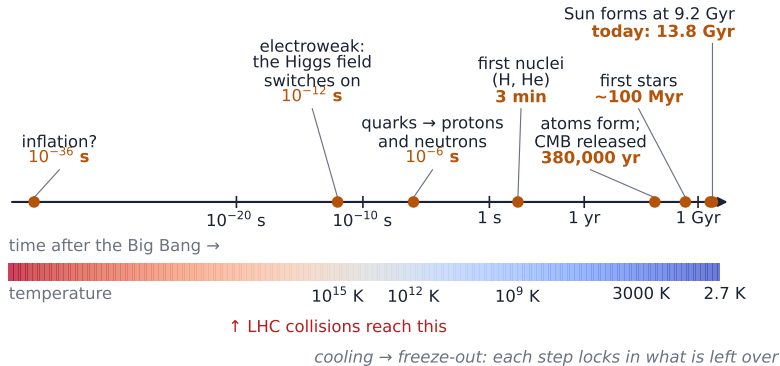
- ▶ Light takes time to travel: the Sun as it was 8 minutes ago, Andromeda 2.5 million years ago
- ▶ The farthest galaxies (JWST) we see as they were **13+ billion years ago**
- ▶ Telescopes are time machines: look far enough and you see the young universe

The universe is expanding



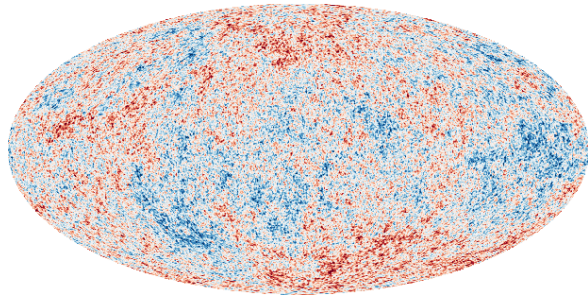
- ▶ Distant galaxies recede, faster the farther they are: $v = H_0 d$ (Hubble, 1929)
- ▶ Not flying apart *through* space: **space itself stretches** (a balloon being inflated)
- ▶ Light is stretched too: **redshift**

Run the film backwards: a hot beginning



- ▶ Expanding means cooling. Going back: hotter and denser, until everything was particles
- ▶ As it cooled, structures **froze out** in order: protons, nuclei (3 min), atoms (380,000 yr)

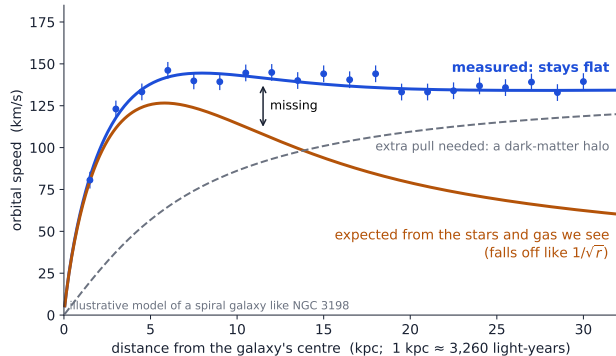
The oldest light: the cosmic microwave background



-300 -150 0 150 300
temperature difference from the 2.7255 K average (μK)
Simulated sky drawn from the Planck 2018 power spectrum (ESA/Planck)

- ▶ The light released at 380,000 years, now stretched to microwaves (2.7 K)
- ▶ Same in every direction to **1 part in 100,000**; the ripples are the seeds of galaxies

What we don't understand: dark matter

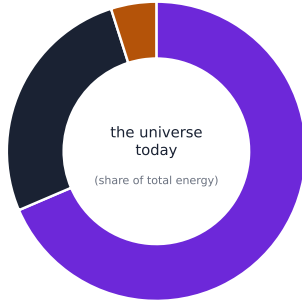


- ▶ Galaxies spin too fast for the matter we can see; clusters hold together too tightly
- ▶ Gravitational lensing, the Bullet Cluster, the CMB all agree: $\sim 5\times$ more **dark matter** than ordinary matter
- ▶ It's the **scaffolding** galaxies formed on. We don't know what it is

Dark energy

- ▶ Einstein added a **cosmological constant** to keep the universe static; called it his biggest blunder when expansion was found
- ▶ 1998: distant supernovae are **fainter** than expected — the expansion is **accelerating**
- ▶ Something with negative pressure fills space: **dark energy**. Einstein's constant, revived

We understand about 5%



68.5% dark energy

drives the accelerating expansion

26.5% dark matter

gravitates, but emits no light

4.9% ordinary matter

atoms: mostly gas between galaxies;
stars are only about 0.3%

Planck 2018 (ESA/Planck); neutrinos and light $\approx 0.1\%$

- ▶ Everything in the Standard Model is the 5%
- ▶ **Inflation:** an early burst of expansion. Consistent with the data, **not confirmed**

Recap of Part 1

- ▶ Data science needs a domain; you'll practise on mine
- ▶ To see small, go to high energy; collisions also **make** new particles
- ▶ The Standard Model: 12 matter particles, force carriers, the Higgs
- ▶ The universe began hot; colliders recreate it
- ▶ 95% of it is dark matter and dark energy, and we don't know what they are

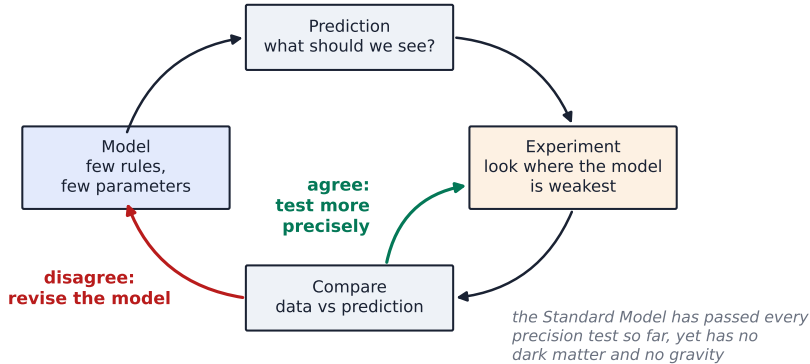
Part 2: why build the LHC, and what it takes to see a collision

Part 2 — Why build the LHC, and how

Was the universe an accident?

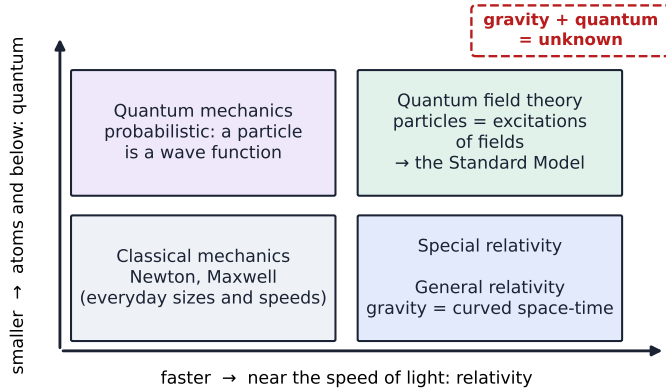
- ▶ Change the constants of nature a little and there are no stars, no chemistry, no us
 - ▶ Gravity's strength, the vacuum energy (off by ~ 120 orders of magnitude), the Higgs mass
 - ▶ The universe began in an extraordinarily ordered state: 1 in $10^{10^{123}}$ (Penrose)
- ▶ Three kinds of answer
 - ▶ **Chance:** many universes; we find ourselves in one that allows us
 - ▶ **Naturalness:** a deeper theory makes these numbers inevitable
 - ▶ **Design:** outside science

Science is a loop



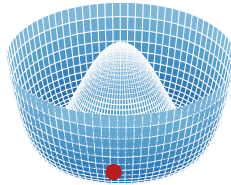
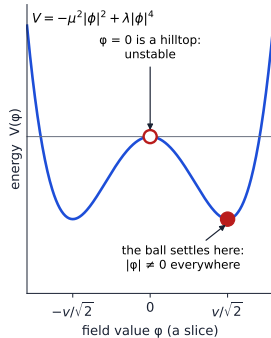
- Experimenters' job: **shoot down theories**. A model survives until data disagree

Our mathematical models



- **Quantum field theory:** where small and fast meet. A prescription for computing, not a picture

The Standard Model's recipe



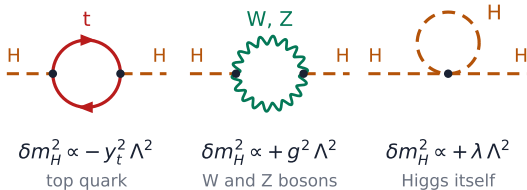
- ▶ **Particles + symmetries** (which require the force carriers)
- ▶ The **Higgs field** fills space; particles moving through it get mass
- ▶ Found at the LHC in 2012 at 125 GeV; Nobel prize 2013

What the Standard Model leaves out

- ▶ 19 numbers it can't predict (26 once neutrinos have mass): masses, mixings, strengths
- ▶ **Gravity**
- ▶ **Dark matter** and **dark energy**
- ▶ Why **matter** beat antimatter
- ▶ Why the Higgs is **so light** (next slide)

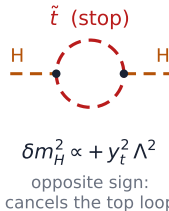
Box: why the Higgs mass is a puzzle

Standard Model



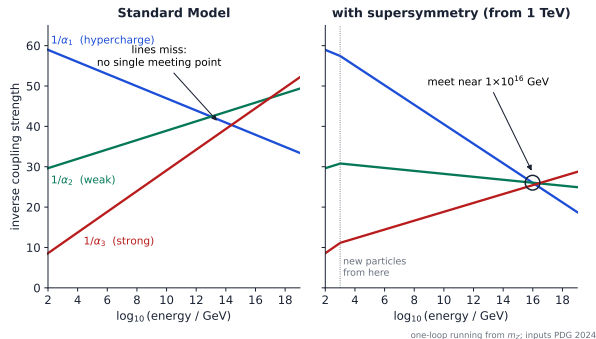
Λ : the highest energy the theory holds to

supersymmetry adds



- ▶ The Higgs mass picks up corrections from every particle it touches:
 $m_H^2 = m_0^2 + \delta m^2$
- ▶ δm^2 grows with the highest energy the theory holds to. If that's 10^{16} GeV, two huge numbers must cancel in m^2 to about **1 part in 10^{28}** to leave $(125 \text{ GeV})^2$

Box: supersymmetry's answer



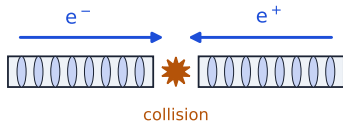
- ▶ Give every particle a partner of the other spin type; their loops **cancel**
- ▶ Bonuses: the three forces **unify** (right), and the lightest partner could be **dark matter**
- ▶ **Not seen.** LHC limits: gluino partners above ~ 2 TeV. Still motivated, increasingly squeezed

Naturalness points to ~ 1 TeV

- ▶ If something fixes the Higgs mass, it should appear near the **TeV scale**
- ▶ That is why the LHC was built to reach a few TeV per collision
- ▶ So far: the Higgs, and nothing else new. Either it's harder to see, higher in energy, or nature isn't "natural"
- ▶ Finding nothing is also data: it rules out whole classes of theories

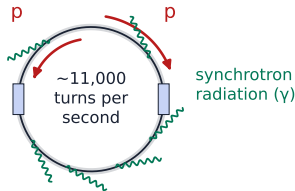
How do you reach a TeV?

linear collider



two accelerators fire at each other;
each bunch gets one pass, so the
energy reached is set by the length

circular collider



$$\text{energy lost per turn} \propto E^4/(m^4 R)$$

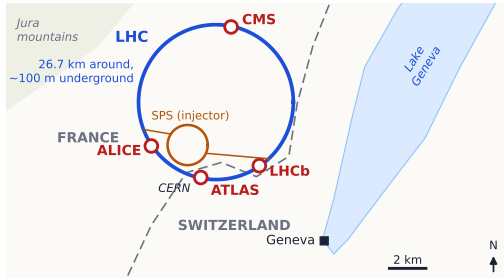
protons are 1836 $\times$ heavier than electrons:
at the same energy they lose
 $(m_p/m_e)^4 \approx 10^{13}$ times less

- ▶ Particles **surf** on radio-frequency waves, gaining energy each pass
- ▶ Circles reuse the accelerator, but bending costs **synchrotron radiation** $\propto (E/m)^4$
- ▶ Protons are $\sim 2000\times$ heavier than electrons: a proton ring can go far higher

The one that got away: the SSC

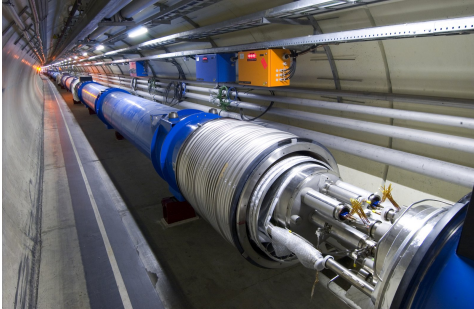
- ▶ The **Superconducting Super Collider**: 87 km ring, 40 TeV, south of Dallas (Waxahachie)
- ▶ Tunnelling started; Congress **cancelled it in 1993**
- ▶ Its people and computers stayed: they helped seed the Dallas–Fort Worth data-centre industry
- ▶ The world went to CERN instead

The Large Hadron Collider



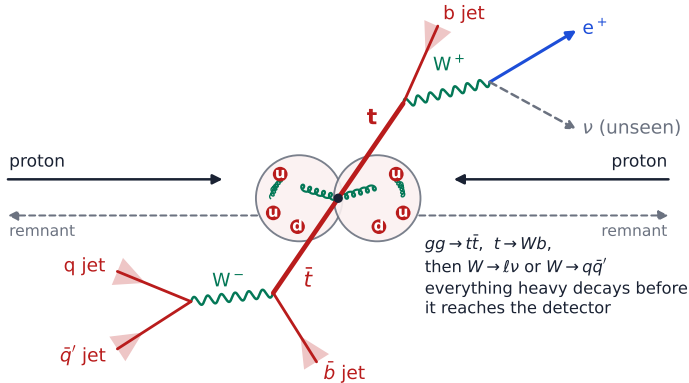
- ▶ 27 km ring, 100 m under the France–Switzerland border · four experiments: **ATLAS**, CMS, ALICE, LHCb

The LHC by the numbers



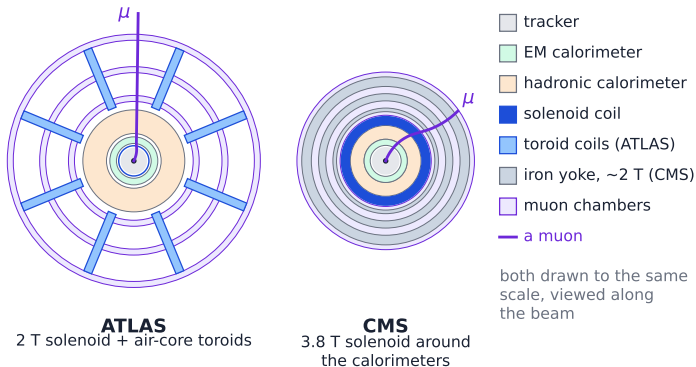
- ▶ 6.8 TeV per beam → **13.6 TeV** collisions (Run 3, since 2022)
- ▶ 1232 superconducting dipoles, 8.3 T, at **1.9 K** — colder than outer space
- ▶ Up to ~ 2800 bunches of $\sim 10^{11}$ protons, crossing every **25 ns = 40 MHz**
- ▶ ~ 60 collisions per crossing (**pile-up**)

Protons are bags of quarks



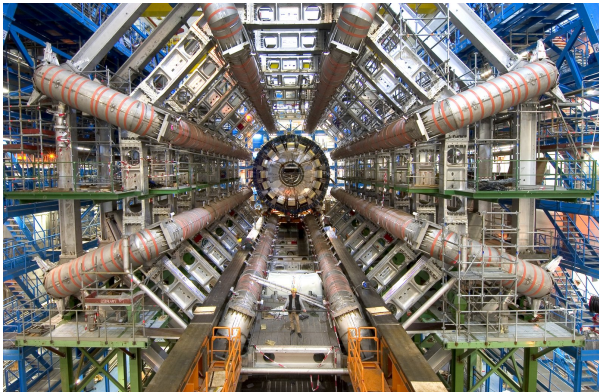
- ▶ A collision is between **one piece of each** proton: you don't know its energy in advance
- ▶ What's left of the protons flies down the beam pipe, unseen
- ▶ Heavy things made (top quarks, W, Higgs) **decay** before they can be seen

Two detectors, two designs



- ▶ ATLAS and CMS measure the same things with different magnets — and **don't share data**
- ▶ A discovery needs both to see it: **independent replication**

ATLAS



- ▶ 46 m long, 25 m high, 7000 tonnes; ~100 million readout channels
- ▶ ~6000 collaboration members from ~180 institutions — UTA among them

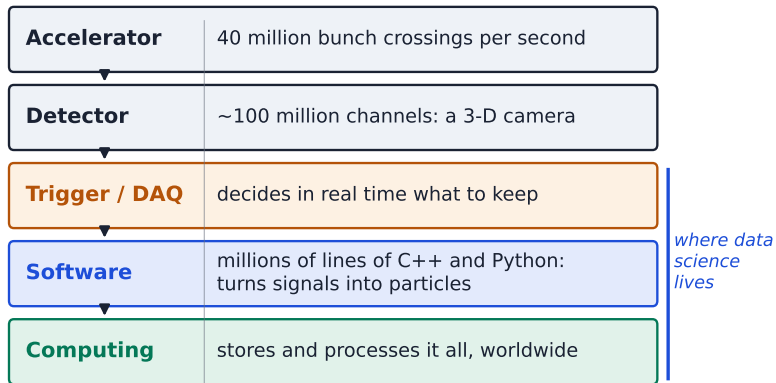
Recap of Part 2

- ▶ The Standard Model works, and can't be complete
- ▶ Naturalness said: new physics near 1 TeV. The LHC was built to look
- ▶ 13.6 TeV proton collisions, 40 million times a second
- ▶ ATLAS and CMS: independent, competing cameras

Part 3: what a detector actually records, and how a trillion pictures become a discovery

Part 3 — From collisions to data

An experiment has five parts

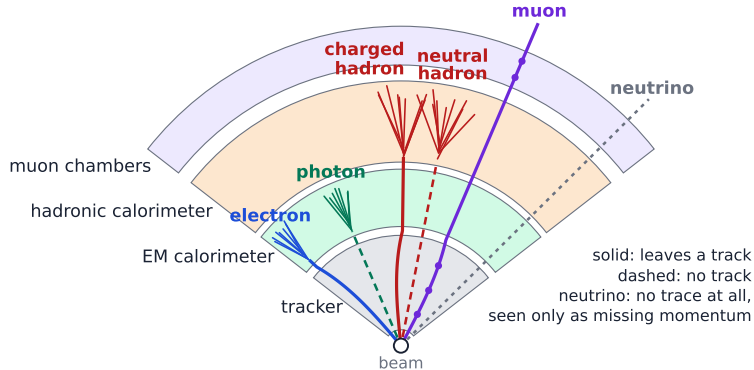


► Each layer is an effort as big as the others

A camera with 100 million pixels

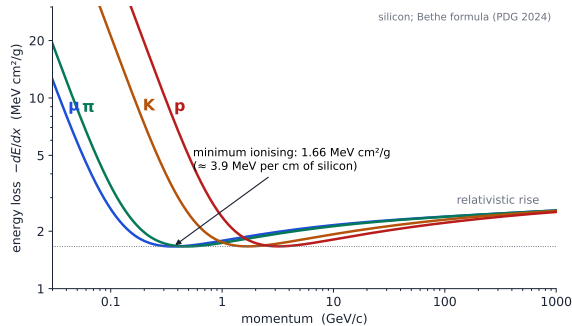
- ▶ Each crossing: a 3-D picture with ~100 million readout channels
- ▶ **40 million pictures a second**
- ▶ The same silicon pixel technology is now in your phone camera
- ▶ Most of the work: turning pixels into **particles** (*reconstruction*)

What survives long enough to see



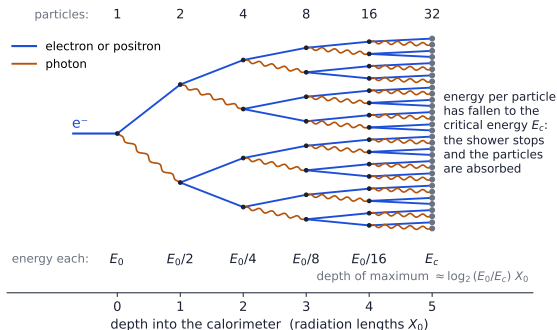
- ▶ Only a handful of particles live long enough: e , γ , μ , charged hadrons (π , K , p), neutral hadrons (n), ν
- ▶ Each leaves a different **signature** across the layers. Neutrinos leave nothing

Tracking



- ▶ Charged particles **ionise** the material they cross: a trail of hits
- ▶ In a magnetic field they curve: $p_T [\text{GeV}] = 0.3 B [\text{T}] R [\text{m}]$
- ▶ How much energy they deposit (dE/dx) depends on speed: it can tell particles apart

Calorimetry

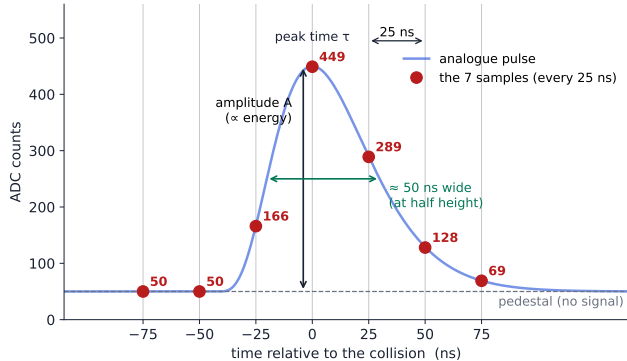


- ▶ Stop the particle in dense material; it **showers** into many particles; count their energy
- ▶ EM calorimeters: high-Z absorbers (lead, tungsten). Hadronic: thick steel, copper
- ▶ Resolution improves with energy: $\sigma/E \approx a/\sqrt{E} \oplus c$

Muons and missing momentum

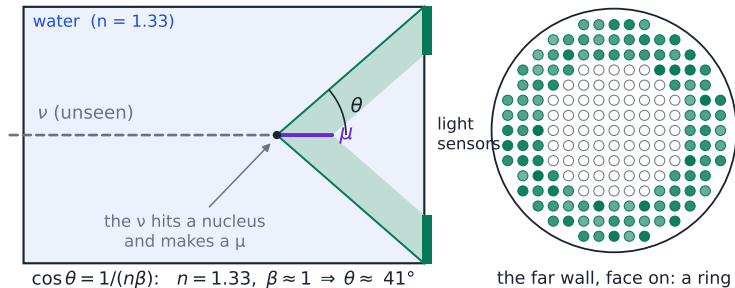
- ▶ **Muons** pass through everything; the outermost layer is a muon spectrometer
- ▶ **Neutrinos** escape. Sideways momentum starts at zero, so it must sum to zero: whatever is missing is **missing transverse momentum** (E_T^{miss})
- ▶ Missing momentum is also how **dark matter** would show up

A real signal: the Tile calorimeter



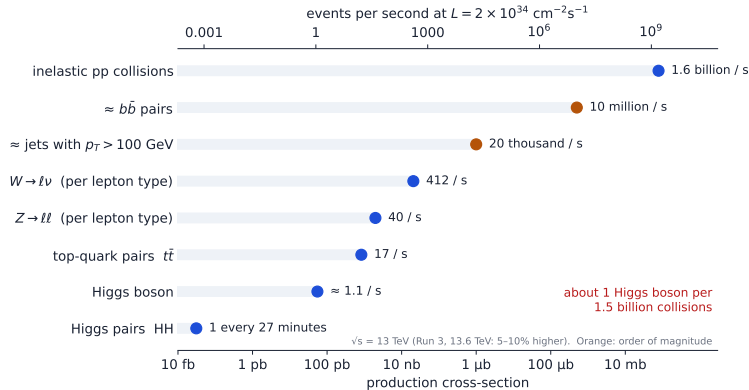
- ▶ Steel plates and plastic tiles; light carried by fibres to photomultipliers
- ▶ Each cell reports **7 numbers every 25 ns**. From those: energy and time
- ▶ This is where the data begins: **ADC counts**, not particles

Box: other ways to see particles



- ▶ Faster than light *in water* $\rightarrow$ a cone of light, a “light boom” (Super-Kamiokande, neutrinos)
- ▶ Liquid-argon time-projection chambers: 3-D photographs of neutrino interactions (DUNE)

A needle in a haystack



- ▶ ~ 1 billion collisions a second; ~ 1 **Higgs** a second; 1 in 10^{12} a Higgs $\rightarrow \gamma\gamma$
- ▶ Every picture is different: nature “rolls dice” each collision

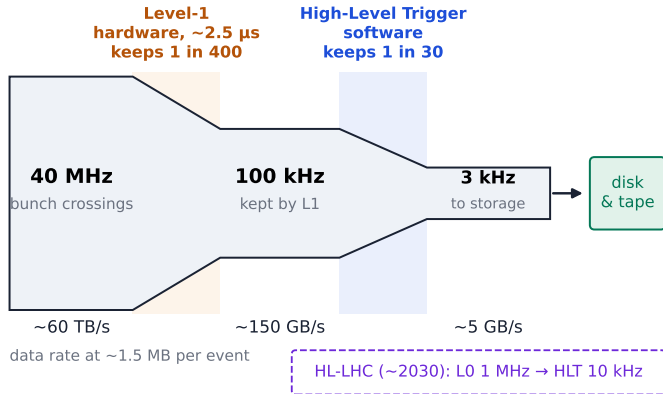
What “background” means

- ▶ **Signal:** the rare process you're looking for
- ▶ **Background:** common processes that *don't really look like* the signal
 - ▶ but there are so many of them that once in a while one does
 - ▶ and those few outnumber the signal
- ▶ Rare signal, overwhelming background: this sentence is the rest of the course

Back of the envelope

- ▶ $40 \text{ MHz} \times \sim 1.5 \text{ MB per picture} \approx \mathbf{60 \text{ TB per second}}$
- ▶ Storing it: **2 disks of 30 TB every second**, $\sim 2 \times 10^7$ disks a beam year
- ▶ Just moving it: $\sim 5,000$ fibres at 100 Gb/s each
- ▶ So: **decide in real time which pictures to keep**

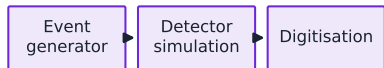
The trigger



- Keep ~ 1 crossing in 10,000; everything else is gone forever

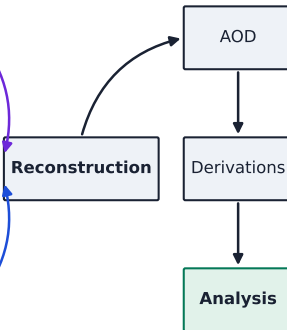
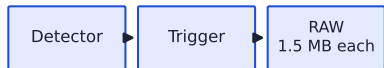
Simulation, reconstruction, analysis

Simulation



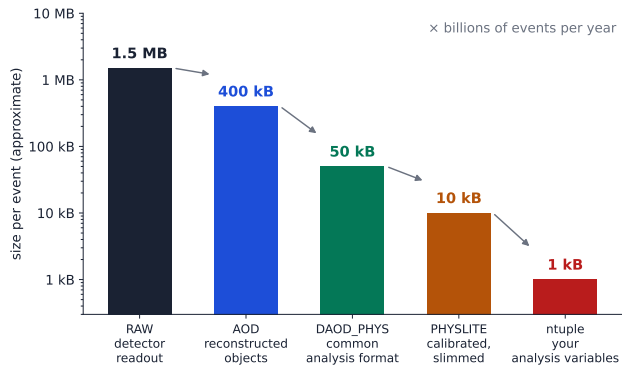
*the same software processes both:
design the analysis on simulation,
apply it once to data*

Real data



- ▶ Simulate everything first: where you know the answer, design the analysis
- ▶ Then apply it **once** to the real data

Analysis is mostly reduction



- ▶ At each step: make decisions, keep the results, discard the inputs
- ▶ 1% of storage or CPU costs **tens of millions of dollars**: formats matter

The grid

~1.5 million CPU cores · over an exabyte stored

Tier-0 (CERN)

records RAW, first reconstruction

Tier-1 (~13)

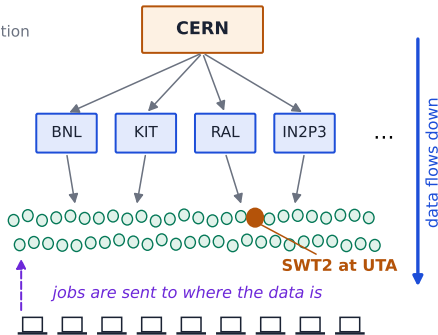
national centres:
copy of RAW, tape

Tier-2 (~150+)

universities and labs:
simulation, analysis

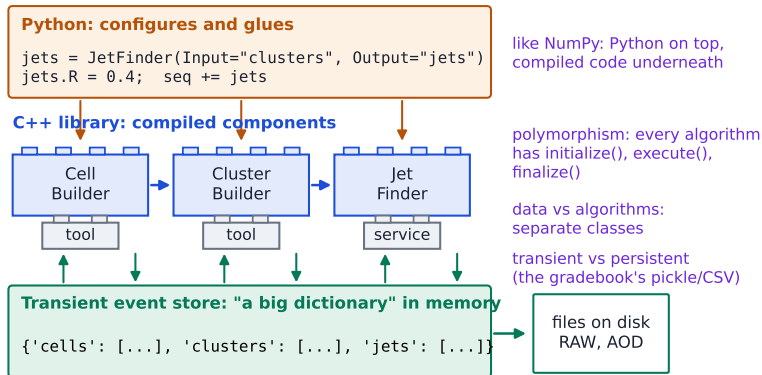
Tier-3 / users

your laptop or local cluster



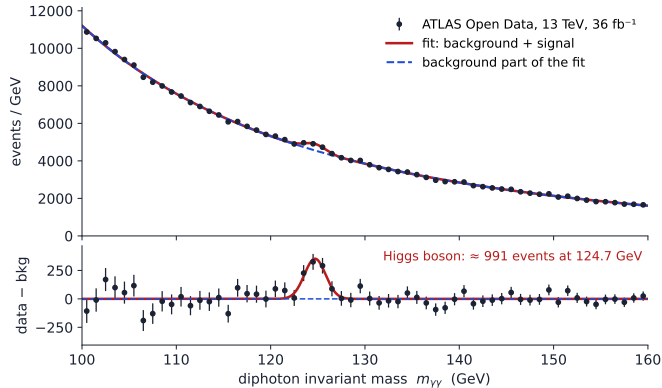
- ▶ Hundreds of sites, ~1.5 million cores, more than an exabyte stored
- ▶ The job-management system ATLAS runs worldwide (PanDA) was developed partly at UTA

Software as Lego



- ▶ Chapter 4's ideas at scale: data separate from algorithms, components behind a common interface, transient vs persistent storage
- ▶ Python glues compiled pieces together — the same model as NumPy

Anatomy of a discovery: $H \rightarrow \gamma\gamma$

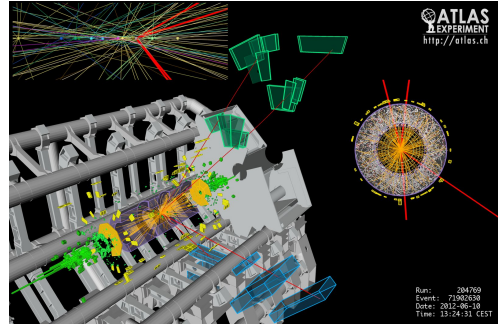
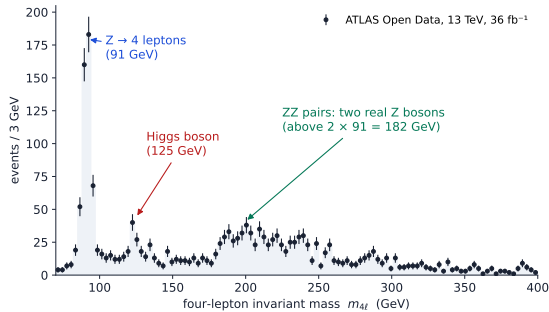


- ▶ Add the two photons' four-vectors, compute the mass (Chapter 4, Lecture 9) — histogram it
- ▶ Background falls smoothly; a Higgs makes a **bump** at 125 GeV

Five sigma

- ▶ Is the bump real, or a fluctuation? Count: background B , excess S , size $\approx S/\sqrt{B}$
- ▶ 5σ : a fluctuation that large happens about **1 in 3.5 million** times
 - ▶ a convention from experience, not a law of nature
- ▶ **Blind analysis**: decide the method before looking
- ▶ 4 July 2012: ATLAS and CMS, independently, both above 5σ

A second look: four leptons



- ▶ $H \rightarrow ZZ \rightarrow 4 \text{ leptons}$: very little background, very few events
- ▶ Right: one such collision as ATLAS saw it

Recap of Part 3

- ▶ A detector is a camera; reconstruction turns pixels into particles
- ▶ Rare signal, overwhelming background: a **trigger** keeps 1 in 10,000 pictures
- ▶ Simulate to design; reduce at every step; apply once, blind
- ▶ A discovery is a histogram and a count

Next (Lab 7 / Chapter 8): your turn — one row per collision, a truth label, and a 2 GB file